

Architectural Inductive Biases Shape Chess Intelligence: CNNs Learn Tactics, Transformers Learn Strategy

Rami Abu Mukh¹, Omar Garah¹, and Ameer Khalaily¹

No Institute Given

Abstract. We present a controlled comparison of CNN and Transformer architectures for chess move prediction, training both identically on 13.3M positions. Despite higher validation accuracy (52.1% vs 50.8%), Transformer exhibits different understanding. Analysis of 128 head-to-head games, 1M tactical puzzles, and phase-specific play reveals architectural specialization: CNNs develop tactical expertise, excelling at local patterns (back-rank mates, forks) through spatial inductive bias, while Transformers develop strategic sophistication, dominating openings and complex positions via global attention. Transformer dominates direct matches (48.4% vs 32.8%), while puzzle performance is nearly identical (38.07% vs 37.91%) despite distinct theme specialization. Phase analysis shows striking divergence: Transformers excel in openings (95.8%) but collapse in endgames (53.9%), while CNNs maintain consistent precision (84-85%) across all phases. Architectural choice determines the type of chess intelligence learned: tactical calculation versus positional understanding, not just performance level.

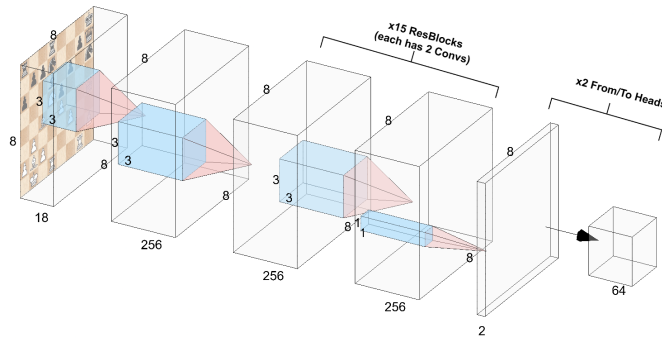


Fig. 1. CNN-based policy architecture used in this study. The input board tensor (18 channels, 8×8 spatial grid) passes through an initial convolution, 15 deep residual blocks, and two separate convolutional heads for from-square and to-square prediction. This spatial structure enables efficient recognition of local tactical patterns.

1 Introduction

Deep neural networks have revolutionized chess engines since AlphaZero’s breakthrough demonstration of superhuman play through pure self-play reinforcement learning [1]. AlphaZero’s success relied on a deep residual CNN to evaluate positions and generate move probabilities, establishing convolutional architectures as the dominant paradigm for learned chess engines [2]. However, CNNs’ localized receptive fields may limit their ability to capture long-range piece interactions and global board patterns, a limitation noted by the AlphaZero authors themselves [1].

Transformers, introduced by Vaswani et al. [3], offer a compelling alternative through their global self-attention mechanism, which directly models relationships across the entire input without distance constraints. Recent works have begun exploring transformers for chess: Noever et al. [4] demonstrated that GPT-2-based models can generate legal chess moves, while Monroe et al. [5] showed their “Chessformer” architecture could match or exceed CNN-based engines. Despite these promising results, prior comparisons have varied training regimes, datasets, and model scales, making it difficult to isolate the effect of architecture alone.

Our contribution. We present the first rigorously controlled comparison of CNN versus Transformer architectures for chess move prediction. Both models have comparable capacity (~ 18 M parameters) and are trained from scratch on the same 13.3M position dataset with identical hyperparameters, optimization, and hardware. We evaluate on three complementary benchmarks: (1) validation accuracy on expert games, (2) tactical puzzle solving (1M puzzles), and (3) real gameplay on Lichess.org.

Key findings. CNNs develop tactical expertise (local pattern recognition, geometric calculation) while Transformers develop strategic sophistication (opening preparation, non-local coordination). Despite nearly identical validation accuracy (52.1% vs 50.8%), the architectures exhibit stark behavioral specialization: CNNs dominate local geometric puzzles while Transformers win head-to-head matches decisively. These differences demonstrate that architectural choice determines the type of chess intelligence learned, not merely performance level.

2 Background

2.1 CNNs for Chess: Spatial Inductive Bias

Convolutional neural networks treat chess positions as 2D spatial grids, naturally aligning with the board’s geometric structure. CNNs excel at chess through two key properties: (1) *translation equivariance*, tactical patterns like forks or pins are detected by the same filters regardless of board location, and (2) *local-to-global processing*, stacking convolutional layers progressively expands receptive fields from local piece interactions to whole-board patterns. This spatial bias makes CNNs highly efficient at learning tactical motifs, as evidenced by AlphaZero [1] and Leela Chess Zero [6].

However, CNNs’ reliance on localized filters can limit their ability to model long-range dependencies. A CNN requires many layers before distant pieces can directly interact, potentially hindering the representation of strategic concepts involving global coordination (e.g., king safety with pieces spread across the board). This architectural constraint motivated exploring alternatives.

2.2 Transformers for Chess: Global Attention

Transformers process positions as sequences of tokens with self-attention, enabling any position to directly interact with any other from the first layer [3]. This global receptive field theoretically facilitates modeling non-local dependencies, distant piece coordination, complex strategic plans, without requiring deep layer stacks.

The trade-off: transformers lack inherent spatial structure. While CNNs ”know” adjacent squares are related through convolutional kernels, transformers must learn all spatial relationships from data. This flexibility enables expressiveness but sacrifices sample efficiency for inherently spatial tasks. Our controlled comparison isolates whether this trade-off favors CNNs’ efficiency or Transformers’ flexibility for chess.

3 Methodology

3.1 Experimental Design

We frame chess move prediction as supervised learning: predict the move played by an expert given a position. Rather than predicting over all possible moves (~ 4000 classes), we use a factorized approach with two simultaneous 64-way classifications: (1) *from-square* (piece to move) and (2) *to-square* (destination). A move is correct only if both predictions match the ground truth. We train with cross-entropy loss and label smoothing ($\epsilon = 0.1$) [1].

Dataset. We use the Lichess Elite 2022 dataset (LE22ct)[7]: 13.3M positions from games between 2400+ Elo players. The dataset is split 90/10 for training/validation. Both models see identical training examples in the same order (fixed random seed), ensuring differences arise solely from architecture, not data variance.

3.2 Model Architectures

CNN Policy Network. Our ResNet-style CNN (Figure 1) processes positions as $18 \times 8 \times 8$ tensors (12 piece channels + 6 metadata channels). The architecture comprises: (1) initial 3×3 convolution to 256 filters, (2) 15 residual blocks with batch normalization and dropout, and (3) dual 1×1 convolutional heads producing 64-dimensional from/to logits. Total parameters: 17.8M. See Appendix A for detailed specifications.

Transformer Policy Network. Our encoder-only Transformer (Figure 2) processes positions as 69-token sequences (1 turn + 4 castling + 64 board

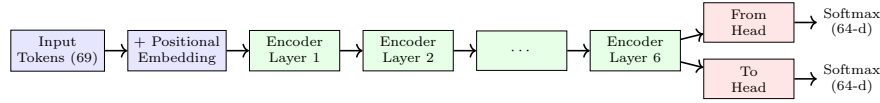


Fig. 2. Transformer architecture: tokens pass through embeddings and 6 encoder layers before projection to from/to distributions.

squares). Tokens are embedded to 512 dimensions with learned positional encodings, then passed through 6 standard Transformer layers (8-head attention, 2048-dimensional FFN). Linear projection heads map the 64 square representations to from/to logits. Total parameters: 19.0M. See Appendix A for full architecture.

3.3 Training Protocol

Both models trained for 55,000 steps (batch size 2048, ~ 9.4 epochs) using Adam optimizer with Vaswani learning rate schedule [3]. Training used bfloat16 mixed precision on a single NVIDIA H100 80GB GPU. Key difference: the CNN trained in 5.3 hours versus 52 hours for the Transformer due to attention’s quadratic complexity. Neither model showed overfitting (validation tracked training). Full hyperparameters in Appendix B.

3.4 Evaluation Benchmarks

We evaluate on three complementary dimensions:

- (1) **Validation Accuracy:** Top-1 accuracy on 1.33M held-out expert positions, measuring ability to predict master-level moves.
- (2) **Tactical Puzzles:** Accuracy on 1M Lichess puzzles spanning ratings 400-3000 and 50+ tactical themes (forks, pins, mates, etc.), testing pure pattern recognition.
- (3) **Live Gameplay:** Performance of deployed bots on Lichess.org in Bullet (1+0) and Blitz (3+0) time controls, evaluating practical playing strength.

4 Results: Behavioral Differences and Performance

4.1 Training Dynamics and Validation Performance

Figure 3 shows training progression over 55,000 steps. Both models improved rapidly during warmup (0-10K steps), then plateaued by 40-50K steps. The Transformer maintained a consistent 1-2% advantage, achieving 52.1% final move accuracy versus the CNN’s 50.8%. Notably, the CNN plateaued around 40K steps while the Transformer continued improving, suggesting it could benefit from extended training.

Both models found from-square prediction easier than to-square (Table 1). At 55K steps, the CNN achieved 64.2% from-square and 55.6% to-square accuracy;

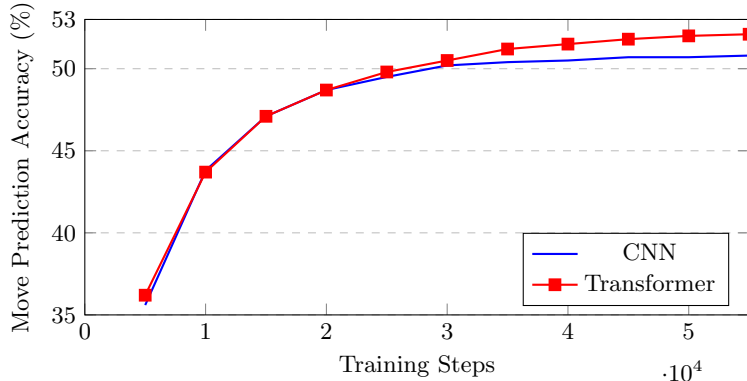


Fig. 3. Validation accuracy progression over 55,000 training steps. Transformer achieves higher final accuracy (52.1% vs 50.8%), establishing the performance baseline for behavioral comparison.

Table 1. Final performance and efficiency metrics for CNN vs Transformer. Both models trained identically except architecture.

Metric	CNN	Transformer
Parameters (M)	17.8	19.0
Training time (hours)	5.3	52.0
Inference (ms/position)	6.2	11.9
From-square accuracy	64.2%	65.6%
To-square accuracy	55.6%	57.3%
Full-move accuracy	50.8%	52.1%

the Transformer reached 65.6% and 57.3% respectively. The Transformer’s larger gain on the harder to-square task (+1.7% vs +0.6% on from-square) suggests its global attention better captures destination dependencies.

4.2 Lichess Gameplay Evaluation

We deployed both models as bots on Lichess.org (**AuraChessCNN**, **AuraChessTransformer**), playing hundreds of games in Bullet (1+0) and Blitz (3+0) formats. Table 2 shows the Transformer achieved modestly higher Elo (+11 in Bullet, +61 in Blitz) with stronger win rates, though not dramatically so. Both bots performed competitively at ~ 1500 -1650 Elo range.

Phase-specific analysis. Using Lichess Insights, we analyzed move accuracy across game phases (Table 3). The Transformer excels in openings (95.8% accuracy, 11.6 centipawn loss) where pattern recognition of standard positions dominates. However, its endgame performance collapses dramatically (53.9% accuracy, 62.7 centipawn loss), suggesting difficulty with sparse, calculational

Table 2. Lichess Elo ratings and game outcomes. The Transformer shows slight game-play advantages in fast time controls.

Variant	Bot	Elo	Win%	Draw%	Loss%
Bullet (1+0)	CNN	1638	46.9	19.6	33.5
	Transformer	1649	48.2	16.8	35.0
Blitz (3+0)	CNN	1516	46.0	15.8	38.2
	Transformer	1577	48.5	18.5	33.0

Table 3. Phase-specific accuracy and centipawn loss (ACPL). The CNN shows consistent performance while the Transformer struggles in endgames.

Phase	Accuracy (%)		ACPL (lower better)	
	CNN	Transformer	CNN	Transformer
Opening	93.5	95.8	16.4	11.6
Middlegame	84.5	82.3	43.0	45.2
Endgame	85.3	53.9	43.2	62.7

positions. Conversely, the CNN maintains consistent 84-85% accuracy across all phases, particularly excelling in endgame tactical awareness (95.2% exploitation rate vs Transformer’s 84%).

4.3 Head-to-Head Match Analysis

To directly compare playing strength, we analyzed 128 games where both bots faced each other on Lichess.org in Blitz (3+0) time control. Table 4 summarizes the results, revealing a clear advantage for the Transformer in direct competition.

The Transformer won 48.4% of games compared to the CNN’s 32.8%, with 18.8% draws. This advantage (15.6 percentage point difference) is more pronounced than their overall Elo ratings suggested (+61 Elo in Blitz). Interestingly, the CNN performed slightly better as Black (36.7% win rate) than as White (29.4%), though the Transformer dominated with both colors. The average game length of 92.3 moves indicates these were substantive games, not quick tactical blunders.

This head-to-head performance confirms the Transformer’s practical game-play advantage in the time controls matching their inference speeds, where its superior opening play and middlegame pattern recognition offset its endgame weaknesses. The CNN’s tactical strengths proved insufficient to overcome the Transformer’s strategic advantages in full-game contexts.

4.4 Tactical Puzzle Benchmark

We evaluated both models on 1M Lichess tactical puzzles: positions with single correct forcing moves. This tests pure pattern recognition, unlike validation accuracy where multiple reasonable moves exist.

Table 4. Head-to-head match results from 128 games on Lichess (Blitz 3+0). The Transformer achieves decisive advantage in direct play.

Outcome	Games	Percentage
Transformer wins	62	48.4%
CNN wins	42	32.8%
Draws	24	18.8%
Total games	128	100%
<i>Performance by color:</i>		
CNN as White	20W-9D-39L	29.4% wins
CNN as Black	22W-15D-23L	36.7% wins
Avg game length	92.3 moves	

Table 5. Overall puzzle performance on 1M Lichess puzzles. Aggregate accuracy is nearly identical (0.16% difference), but theme-level analysis reveals architectural specialization.

Metric	CNN	Transformer
Total puzzles evaluated	1,000,000	1,000,000
Correct solutions	380,708	379,148
Accuracy (%)	38.07	37.91
Avg time per puzzle (ms)	6.24	11.86

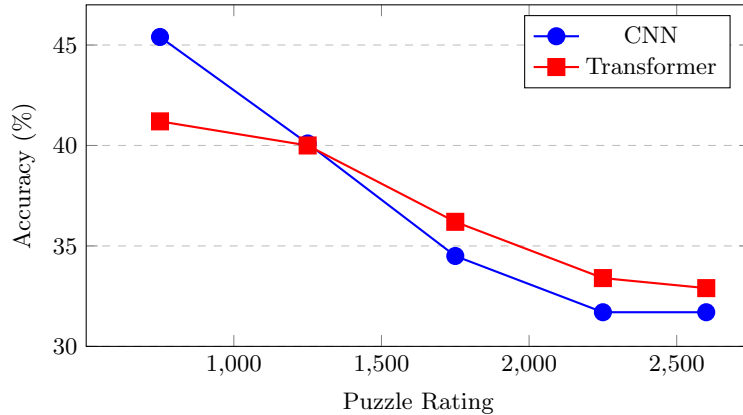
Overall performance. Table 5 shows nearly identical overall accuracy (CNN 38.07%, Transformer 37.91%, a negligible 0.16% difference). This equivalence contrasts with the validation results where the Transformer led by 1.6%, suggesting that raw puzzle-solving accuracy doesn’t reflect the architectures’ distinct behavioral patterns. The true differences emerge at the theme level, not in aggregate performance.

Performance by difficulty. Table 6 and Figure 4 stratify results by puzzle rating, revealing a clear crossover pattern. The CNN dominates easier puzzles (<1500 rating, +4-5% advantage), recognizing simple tactics like hanging pieces and basic forks more reliably. On the hardest puzzles (>2500 rating), the Transformer edges ahead (+1.2%), suggesting its global attention helps with complex multi-piece coordination. The crossover occurs around 1500 rating, indicating task complexity determines optimal architecture choice.

Performance by tactical theme. Table 7 shows accuracy on 15 common themes, revealing clear architectural specialization. Figure 8 visualizes this pattern: the CNN excels at localized tactics: mate-in-1 (+4.5%), mate-in-2 (+2.5%), back-rank mate (+7.0%), and forks (+1.2%). These are patterns involving pieces in direct spatial contact that convolutional filters detect efficiently. Conversely, the Transformer leads on non-local patterns: sacrifices (+2.7%), discovered attacks (+2.7%), attraction (+3.1%), and quiet moves (+1.9%). These require distant piece coordination where global attention provides advantages.

Table 6. Puzzle accuracy by difficulty rating. CNNs excel on simpler tactics; Transformers on complex positions.

Rating	CNN	Transformer	Difference	N puzzles
≤ 1000	45.4%	41.2%	+4.2% (CNN)	212,698
1000–1500	40.1%	40.0%	+0.1% (CNN)	322,425
1500–2000	34.5%	36.2%	+1.7% (Trans)	266,850
2000–2500	31.7%	33.4%	+1.7% (Trans)	158,958
≥ 2500	31.7%	32.9%	+1.2% (Trans)	39,069

**Fig. 4.** Puzzle accuracy by difficulty rating reveals architectural sweet spots. CNN dominates on easier puzzles (≤ 1500 rating), excelling at simple tactics. Transformer gains advantage on harder puzzles (≥ 2000 rating) where complex coordination matters. Lines cross around 1500 rating, indicating task complexity determines optimal architecture choice.

Synthesis. While aggregate puzzle accuracy is nearly identical (0.16% difference), the theme-level analysis reveals profound architectural specialization. CNNs dominate local geometric patterns (+7% back-rank mates, +4.5% mate-in-1) through spatial inductive bias, while Transformers excel at non-local coordination (+3.1% attraction, +2.7% sacrifices/discovered attacks) through global attention. The overall tie masks complementary strengths.

5 Analysis and Interpretation

5.1 Architectural Inductive Biases Shape Chess Understanding

Our controlled experiment reveals that inductive biases fundamentally determine how neural networks learn chess. Despite identical training, the CNN and Transformer develop distinct tactical capabilities aligned with their architectural constraints.

Table 7. Accuracy on top 15 puzzle themes. Positive differences favor CNN (localized tactics); negative favor Transformer (non-local patterns).

Theme	CNN	Transformer	Diff	N puzzles
Mate-in-1	47.7%	43.2%	+4.5%	137,434
Mate-in-2	49.6%	47.1%	+2.5%	129,146
Back-rank mate	54.7%	47.7%	+7.0%	33,515
Fork	37.0%	35.9%	+1.2%	130,086
Pin	36.6%	38.4%	-1.8%	60,713
Skewer	37.7%	39.1%	-1.4%	22,134
Discovered attack	36.8%	39.4%	-2.7%	51,472
Sacrifice	37.9%	40.6%	-2.7%	74,725
Deflection	38.5%	40.0%	-1.5%	43,389
Attraction	36.8%	39.8%	-3.1%	36,053
Quiet move	38.7%	40.6%	-1.9%	41,868
Advantage	31.8%	33.4%	-1.6%	292,281
Endgame	40.8%	39.9%	+1.0%	495,951
Middlegame	35.7%	36.4%	-0.7%	455,285
Opening	32.3%	32.1%	+0.2%	48,764

The CNN’s spatial inductive bias (assuming nearby squares are more related than distant ones) proves decisive for tactical pattern recognition. Its convolutional filters naturally detect local geometric configurations: a king surrounded by enemy pieces (checkmate), knight-fork L-shapes, back-rank mate patterns. The puzzle evaluation confirms this: +7.0% on back-rank mates, +4.5% on mate-in-1, +2.5% on mate-in-2. This tactical precision makes CNNs naturally suited for pattern-based chess understanding.

Conversely, the Transformer’s global attention enables modeling non-local dependencies at the cost of learning spatial relationships from scratch. It excels where distant pieces coordinate: sacrifices (+2.7%), discovered attacks (+2.7%), attraction motifs (+3.1%). On the hardest puzzles (>2500 rating), the Transformer’s flexibility provides +1.2% advantage, suggesting complex positions benefit from unbounded attention. However, this flexibility requires substantially more training time to develop robust spatial understanding.

The phase-specific gameplay analysis crystallizes this divide. The Transformer’s opening strength (95.8% accuracy) leverages pattern recognition of standard positions: its global view integrates castling rights, pawn structure, and development naturally.

The endgame collapse phenomenon. The Transformer’s dramatic endgame failure (53.9% accuracy vs CNN’s 85.3%, a 31.4 point gap) represents the study’s most striking behavioral finding. This collapse reflects a fundamental architectural limitation: endgames require precise geometric calculation (king opposition, square of the pawn, triangulation) where spatial relationships must be computed exactly. The CNN’s convolutional structure encodes spatial geometry directly: “king on e4 is adjacent to d4, e5, f4” is built into the 3×3 kernel structure. The

Table 8. Architectural specialization summary: Key puzzle themes showing complementary strengths. CNNs dominate local geometric tactics; Transformers excel at non-local coordination.

Theme	Difference (CNN - Trans)
<i>CNN advantages (local geometric patterns):</i>	
Back-rank mate	+7.0%
Mate-in-1	+4.5%
Mate-in-2	+2.5%
Fork	+1.2%
<i>Transformer advantages (non-local coordination):</i>	
Attraction	-3.1%
Sacrifice	-2.7%
Discovered attack	-2.7%
Quiet move	-1.9%
<i>Mixed patterns:</i>	
Pin	-1.8%
Skewer	-1.4%

Transformer must learn these relationships from data, and while it succeeds in pattern-rich openings and middlegames, endgame positions are sparser in the training distribution and more diverse in structure. The Transformer’s learned spatial representations fail to generalize to calculational endgames where every square’s geometric relationship matters precisely. The CNN’s consistent 84-85% accuracy across all phases demonstrates its robust spatial reasoning foundation.

5.2 Validation Accuracy Doesn’t Predict Behavioral Patterns

A striking finding: validation accuracy (52.1% vs 50.8% favoring Transformer) doesn’t predict puzzle performance (38.07% vs 37.91%, essentially tied). More importantly, validation accuracy completely fails to predict the dramatic theme-level specialization: CNN +7% on back-rank mates versus Transformer +3% on sacrifices.

Validation positions come from expert games where multiple reasonable moves exist: the "correct" label is merely what a strong player chose. These positions test whether models mimic human move distributions, not whether they understand specific tactical or strategic patterns. Puzzles and head-to-head matches, conversely, test actual chess understanding: puzzles reveal pattern-specific strengths (theme specialization), while matches reveal strategic integration (opening mastery vs endgame calculation).

The CNN and Transformer achieve similar validation accuracy by learning different patterns: CNN focuses on local geometric tactics, Transformer on global strategic themes. Validation accuracy, being an aggregate across diverse positions, masks these complementary specializations. Designers should evaluate on

behaviorally-relevant benchmarks (specific tactical themes, game phases, head-to-head play) rather than aggregate validation metrics.

6 Conclusion

We presented the first rigorously controlled comparison of CNN and Transformer architectures for chess, training both models identically to isolate architectural effects. Our three-pronged evaluation (validation accuracy, tactical puzzles, and live gameplay) reveals nuanced trade-offs that challenge conventional assumptions about architecture selection.

Key finding: Architectural choice determines the type of chess intelligence learned, not merely performance level. CNNs develop tactical expertise (local pattern recognition, geometric calculation) while Transformers develop strategic sophistication (opening theory, non-local coordination). This behavioral specialization explains why the Transformer dominates head-to-head matches (48.4% vs 32.8%), as strategic play matters in full games, while puzzle performance is nearly identical overall (38.07% vs 37.91%) but with dramatic theme-level specialization (CNN +7% on local tactics like back-rank mates, Transformer +3% on non-local patterns like sacrifices).

Architectural specialization. Each architecture develops distinct capabilities aligned with its inductive biases. CNNs’ spatial structure makes them exceptionally efficient at local tactics: +7% on back-rank mates, +4.5% on mate-in-1, +2.5% on mate-in-2. Transformers’ global attention enables non-local reasoning: +2.7-3.1% on sacrifices, discovered attacks, and attraction motifs. Phase analysis confirms this divide: Transformers excel in openings (95.8% accuracy) but collapse in endgames (53.9%), while CNNs maintain consistent 84-85% accuracy across all phases.

A Detailed Architecture Specifications

A.1 CNN Architecture Details

Input Encoding: $18 \times 8 \times 8$ tensor with the following channels:

- Channels 0-5: White pieces (Pawn, Knight, Bishop, Rook, Queen, King)
- Channels 6-11: Black pieces (same order)
- Channel 12: Turn indicator (1.0 for white-to-move, 0.0 for black)
- Channels 13-16: Castling rights (white kingside, white queenside, black kingside, black queenside), broadcast across board
- Channel 17: En passant target square (one-hot encoding)

Network Architecture:

1. Initial convolution: 3×3 kernel, $18 \rightarrow 256$ filters, padding=1
2. Batch normalization + ReLU activation
3. 15 Residual blocks, each containing:

- Conv2d: 3×3 kernel, $256 \rightarrow 256$ filters, padding=1, bias=False
- BatchNorm2d(256) + ReLU
- Conv2d: 3×3 kernel, $256 \rightarrow 256$ filters, padding=1, bias=False
- BatchNorm2d(256) + Dropout2d(p=0.1)
- Skip connection + ReLU
- 4. From-square head:
 - Conv2d: 1×1 kernel, $256 \rightarrow 2$ filters
 - BatchNorm2d(2) + ReLU
 - Select channel 0, flatten to (64,) logits
- 5. To-square head: (identical structure)

Parameter Count: 17,753,096 total parameters

Weight Initialization: Kaiming normal initialization for convolutional layers; constant initialization (weight=1.0, bias=0.0) for batch normalization.

A.2 Transformer Architecture Details

Input Encoding: 69-token sequence:

- Token 0: Turn (0=white, 1=black)
- Tokens 1-4: Castling rights (binary, one per right)
- Tokens 5-68: Board squares a1-h8, encoded as piece type (14 values: empty, 12 piece types, padding)

Network Architecture:

1. Token embeddings: Embedding(vocab_sizes $\rightarrow$ 512) for each token type
2. Positional embeddings: Learned Embedding($69 \rightarrow 512$)
3. Combined embedding scaled by $\sqrt{512}$ + Dropout(0.1)
4. 6 Transformer encoder layers, each containing:
 - Pre-LayerNorm(512)
 - Multi-head self-attention:
 - 8 heads, 64-dimensional queries/keys/values per head
 - Linear projections: $W_Q, W_K, W_V \in \mathbb{R}^{512 \times 512}$
 - Scaled dot-product: $\text{softmax}(QK^T / \sqrt{64})V$
 - Output projection: $W_O \in \mathbb{R}^{512 \times 512}$
 - Dropout(0.1) + residual connection
 - Pre-LayerNorm(512)
 - Position-wise FFN:
 - Linear: $512 \rightarrow 2048$, ReLU, Dropout(0.1)
 - Linear: $2048 \rightarrow 512$, Dropout(0.1)
 - Residual connection
5. Final LayerNorm(512)
6. From-square head: Linear($512 \rightarrow 1$) applied to tokens 5-68, yielding (64,) logits
7. To-square head: (identical structure)

Parameter Count: 18,963,970 total parameters

Weight Initialization: Xavier uniform initialization for linear layers (gain = 1.0); normal initialization for embeddings ($\sigma = 1/\sqrt{512}$).

B Training Hyperparameters

B.1 Optimizer Configuration

- **Optimizer:** Adam
- **Betas:** $\beta_1 = 0.9$, $\beta_2 = 0.98$
- **Epsilon:** 10^{-9}
- **Weight decay:** 0.0 (no L2 regularization)

B.2 Learning Rate Schedule

Vaswani schedule [3]:

$$\text{LR}(\text{step}) = d_{\text{model}}^{-0.5} \cdot \min(\text{step}^{-0.5}, \text{step} \cdot \text{warmup}^{-1.5})$$

CNN: $d_{\text{model}} = 256$, peak LR = 0.0625 at step 8000

Transformer: $d_{\text{model}} = 512$, peak LR = 0.0442 at step 8000

Warmup steps: 8,000 for both models

B.3 Training Configuration

- **Total steps:** 55,000
- **Batch size:** 2,048
- **Total samples seen:** 112,640,000 (~ 9.4 epochs over 11.96M positions)
- **Label smoothing:** $\epsilon = 0.1$
- **Mixed precision:** bfloat16 (BF16)
- **Hardware:** NVIDIA H100 80GB GPU
- **Validation frequency:** Every 5,000 steps
- **Random seed:** 42 (for reproducibility)

C Extended Results

C.1 Training Trajectory Details

Table 9. Training progression showing accuracy and learning rate at key checkpoints.

Step	CNN Acc	Trans Acc	CNN LR	Trans LR	Hours
5,000	35.6%	36.2%	0.00044	0.00031	0.5 / 4.8
10,000	43.8%	43.7%	0.00063	0.00044	1.0 / 9.6
20,000	48.7%	48.7%	0.00044	0.00031	1.9 / 19.1
30,000	50.2%	50.5%	0.00036	0.00026	2.9 / 28.6
40,000	50.5%	51.5%	0.00031	0.00022	3.8 / 37.9
50,000	50.7%	52.0%	0.00028	0.00020	4.8 / 47.2
55,000	50.8%	52.1%	0.00027	0.00019	5.3 / 52.0

C.2 Puzzle Theme Performance

Extended breakdown of all 50+ puzzle themes available upon request. Key patterns:

- **CNN advantages (↓2%):** Smothered mate (+2.2%), Dovetail mate (+0.7%), Hook mate (+0.7%), Back-rank mate (+7.0%), Mate-in-1/2/3 (+2-4.5%)
- **Transformer advantages (↓2%):** Attraction (+3.1%), Discovered attack (+2.7%), Sacrifice (+2.7%), Clearance (+2.3%)
- **Balanced themes (↓1% diff):** Opening (+0.2%), Endgame (+1.0%), Zugzwang (+1.3%)

References

1. David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dhharshan Kumaran, Thore Graepel, Timothy Lillicrap, Karen Simonyan, and Demis Hassabis. A general reinforcement learning algorithm that masters chess, shogi, and go through self-play. *Science*, 362(6419):1140–1144, 2018.
2. David Silver, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George Van Den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, Sander Dieleman, Dominik Grewe, John Nham, Nal Kalchbrenner, Ilya Sutskever, Timothy Lillicrap, Madeleine Leach, Koray Kavukcuoglu, Thore Graepel, and Demis Hassabis. Mastering the game of go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016.
3. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NIPS)*, volume 30, pages 5998–6008, 2017.
4. David Noever, Matt Ciolino, and Josh Kalin. The chess transformer: Mastering play using generative language models. arXiv preprint arXiv:2008.04057, 2020.
5. Daniel Monroe and Philip A. Chalmers. Mastering chess with a transformer model. arXiv preprint arXiv:2409.12272, 2024.
6. Leela Chess Zero Contributors. Leela chess zero. <https://lczero.org>, 2018. Open source neural network based chess engine.
7. Lichess.org. Lichess elite database. <https://database.lichess.org>, 2022. Curated dataset of games from 2400+ rated players.